Fiscal Year:	FY 2009	**Task Last Updated:**	FY 08/31/2009
PI Name:	Hoffman, Jeffrey Ph.D.		
Project Title:	Thermal Control During Astronaut Traverses		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	617-452-2353
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PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02139-4301	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/01/2009	**End Date:**	09/30/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Correct end date is 9/30/2010, per NSBRI (11/3/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Zuber, Maria (Massachusetts Institute of Technology) Newman, Dava (Massachusetts Institute of Technology)		
Grant/Contract No.:	NCC 9-58-HFP00003		
Performance Goal No.:			
Performance Goal Text:			
	NASA’s future plans to explore the Moon and establish a lunar outpost will require extravehicular activities (EVAs) of a quantity and scope never before attempted. Astronauts, robotic assistants and astronaut-carrying vehicles will be used in concert to complete a wide variety of challenging activities. Because of the enormous numbers of EVAs, it will be essential to maximize crew safety and productivity while minimizing the energy cost on the explorers and making most efficient use of limiting consumables like water and electrical energy. We have developed over the past few years a path-planning and navigation tool, called the Mission Planner (Johnson et al. 2009, with references to past work), to assist with pre-mission planning, scenario simulation, real-time navigation and contingency re-planning during planetary traverses. The Mission Planner calculates the most efficient path between user-specified waypoints. Efficiency is based on an exploration cost algorithm, which in the current implementation is		

Task Description:	set equal to the estimated astronaut metabolic rate. We propose to improve our ability to calculate thermal constraints for planetary surface exploration by incorporating solar illumination into the path-planning tool developed over the past few years at MIT. The results will allow modeling thermal stress during on-foot astronaut EVAs as well as during astronaut travel in pressurized rovers. It will also apply to thermal modeling of traverses by robotic exploration vehicles. We will incorporate into our Mission Planner lunar topography and surface roughness data as it becomes available from the Lunar Orbiter Laser Altimeter (LOLA) experiment on the Lunar Reconnaissance Orbiter. Deliverables: • Predict position of the sun for any given time and place on the moon • Calculate sunlit and shadowed area for a given topographical model • Incorporate LOLA altimetry data into topographic model • Calculate thermal loading for sample EVA traverses • Investigate extension of model to small pressurized rovers
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2009.
Bibliography Type:	Description: (Last Updated:)